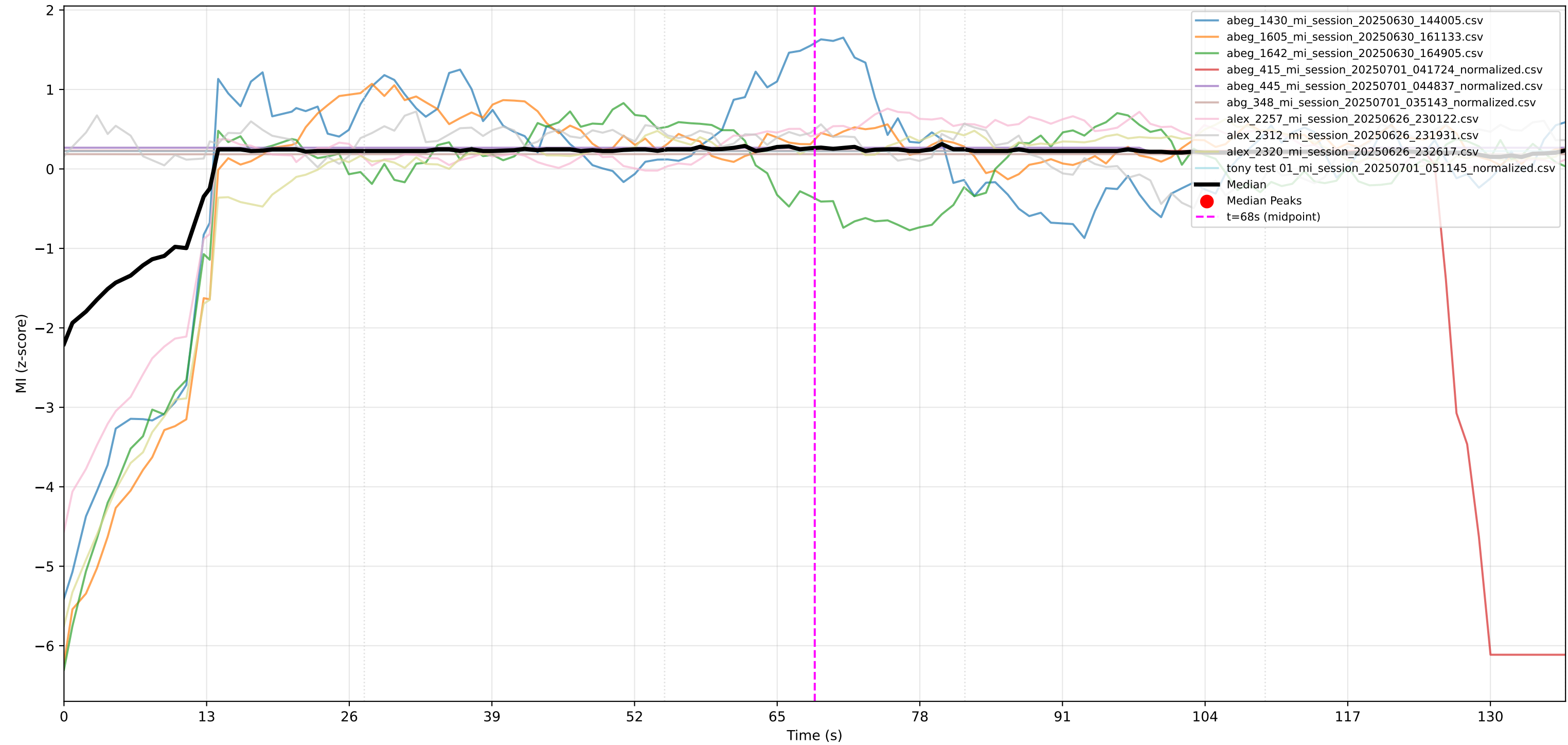
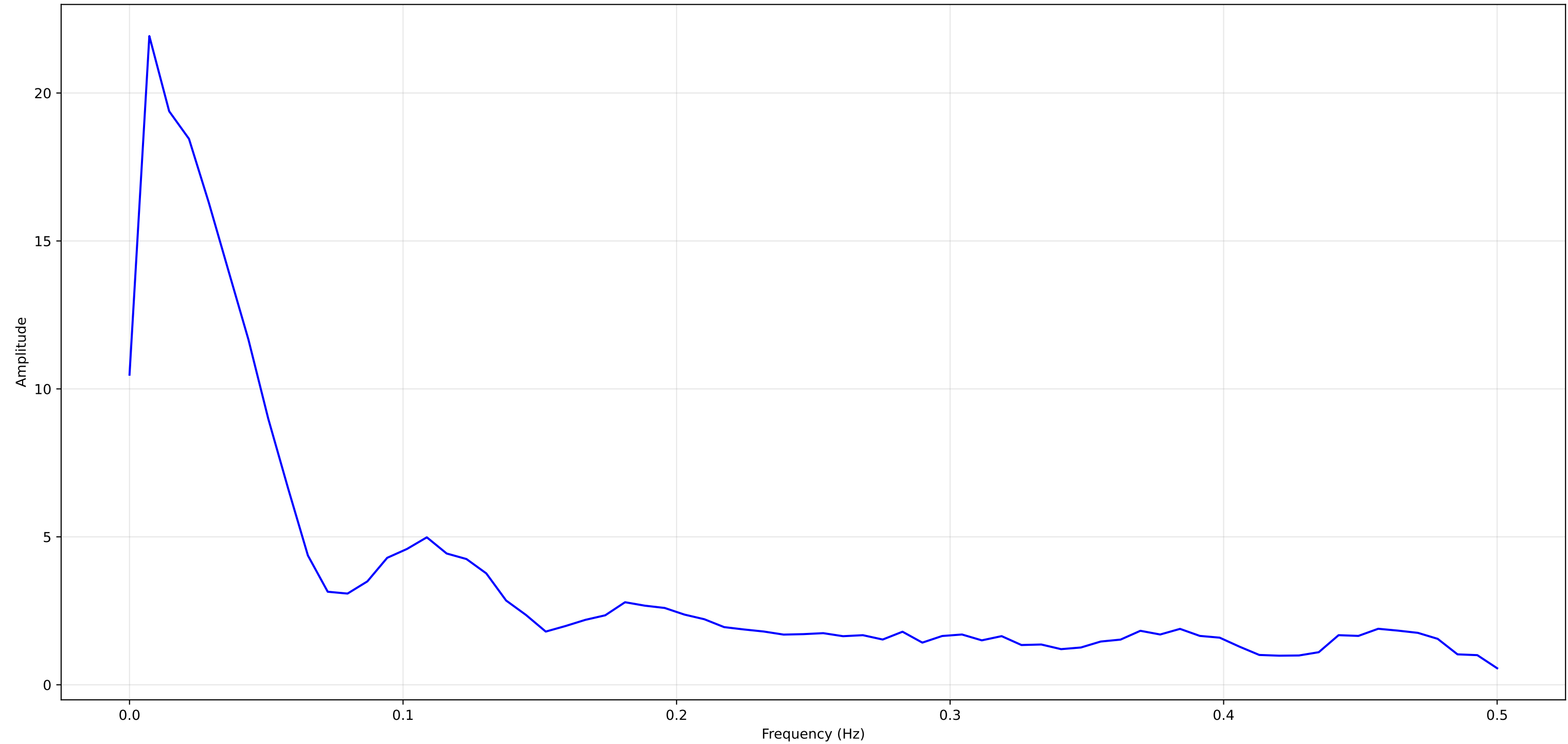


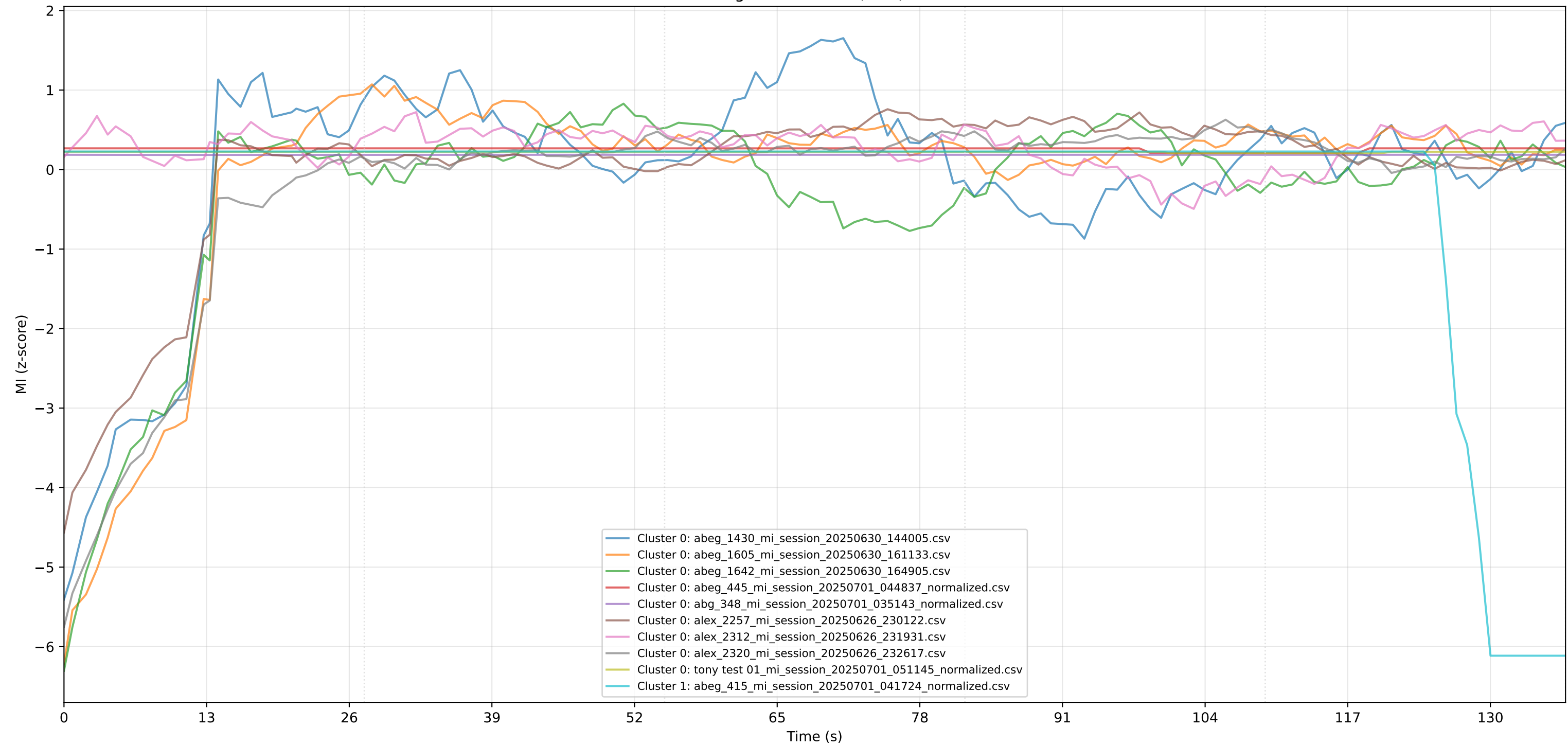
MI Curves (z-score, MA=20) with Median and Peaks (137s Duration)



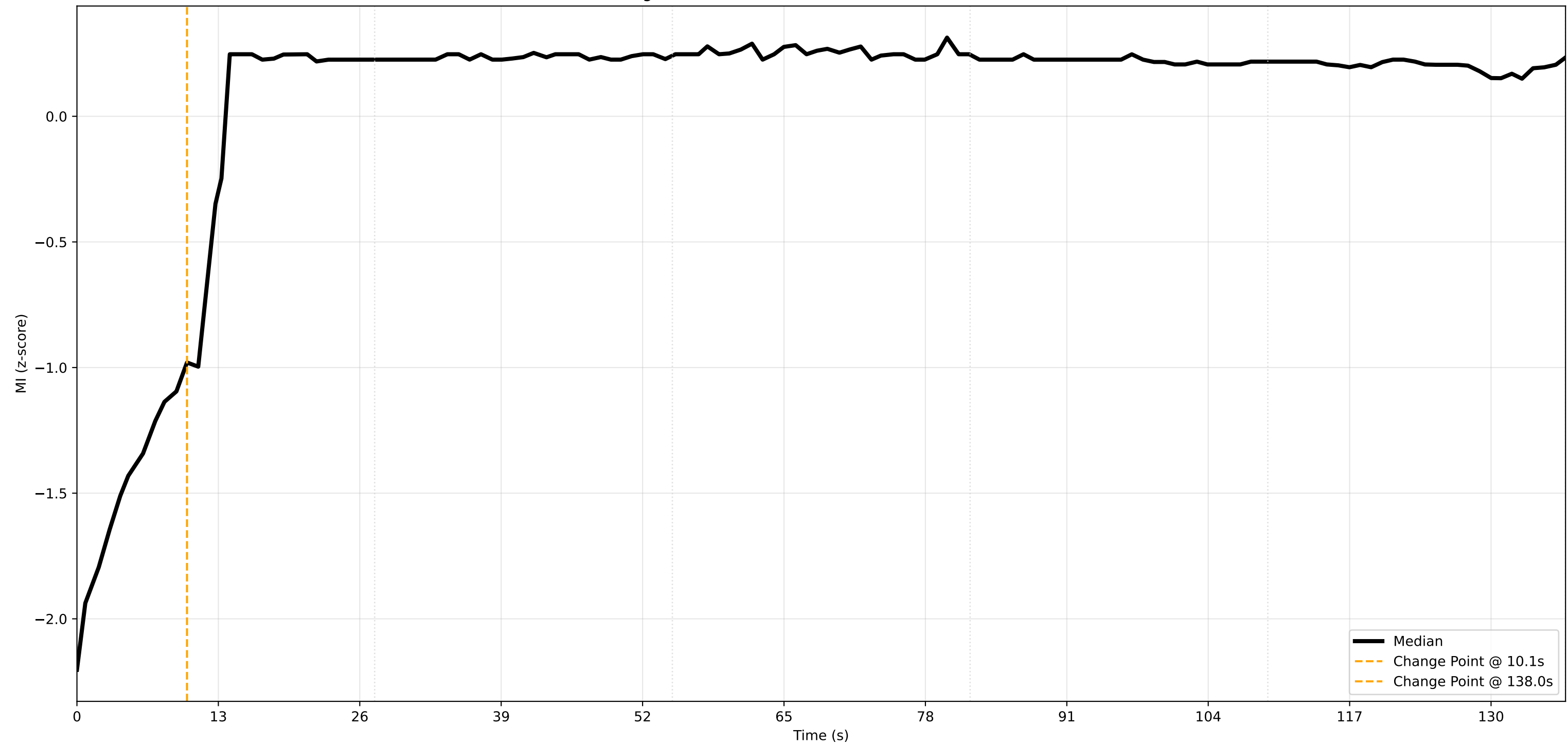
Median MI Frequency Spectrum (FFT)



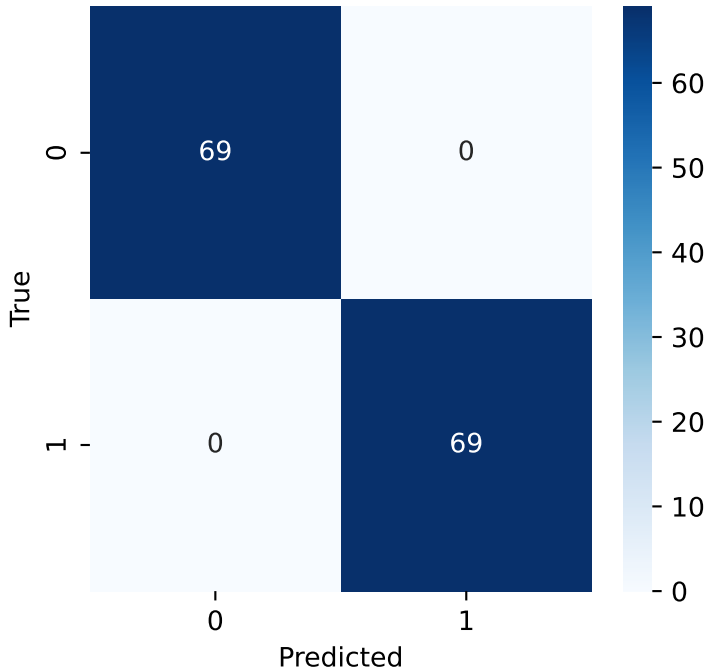
KMeans Clustering of MI Curves (k=2) - 137s Duration



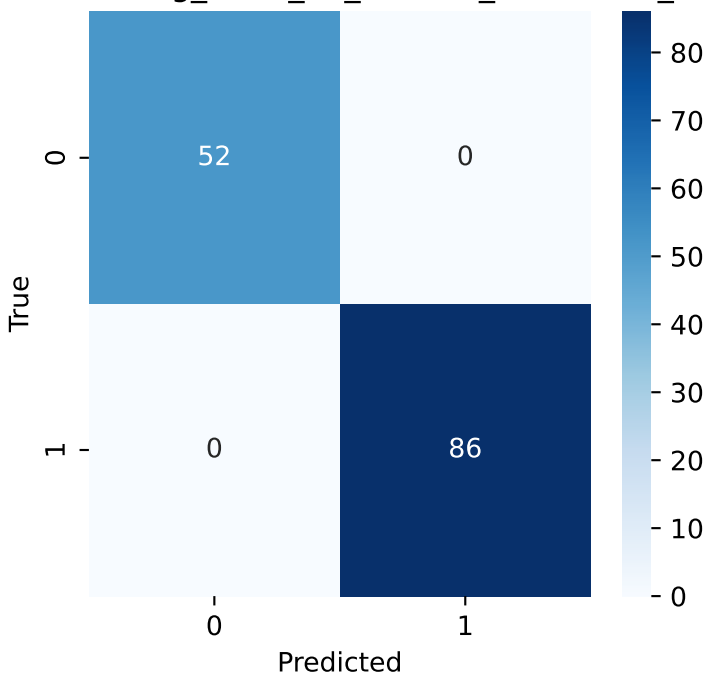
Change Point Detection on Median MI - 137s Duration



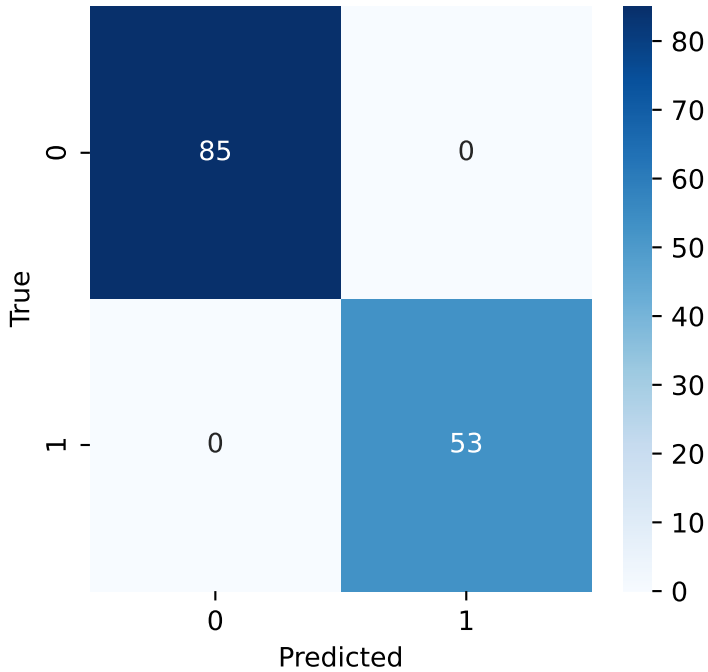
Matrix: abeg_1430_mi_session_20250630_1440



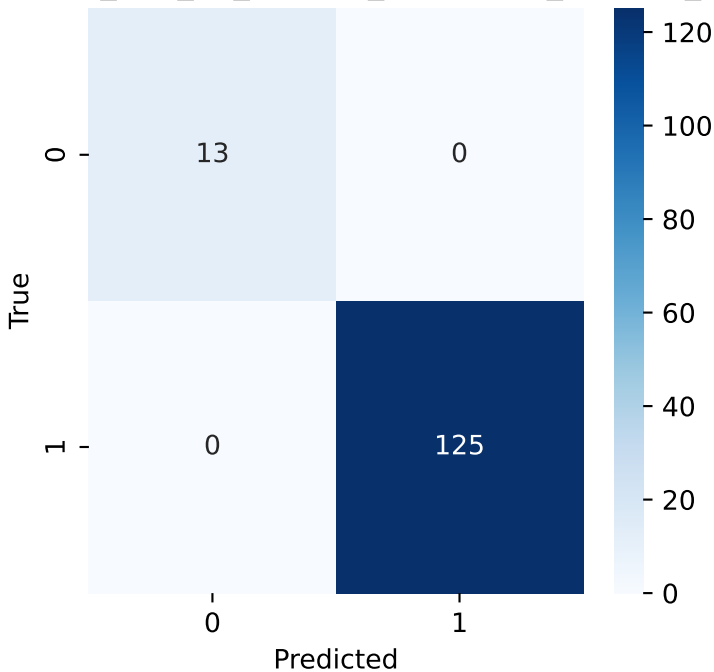
Matrix: abeg_1605_mi_session_20250630_1611



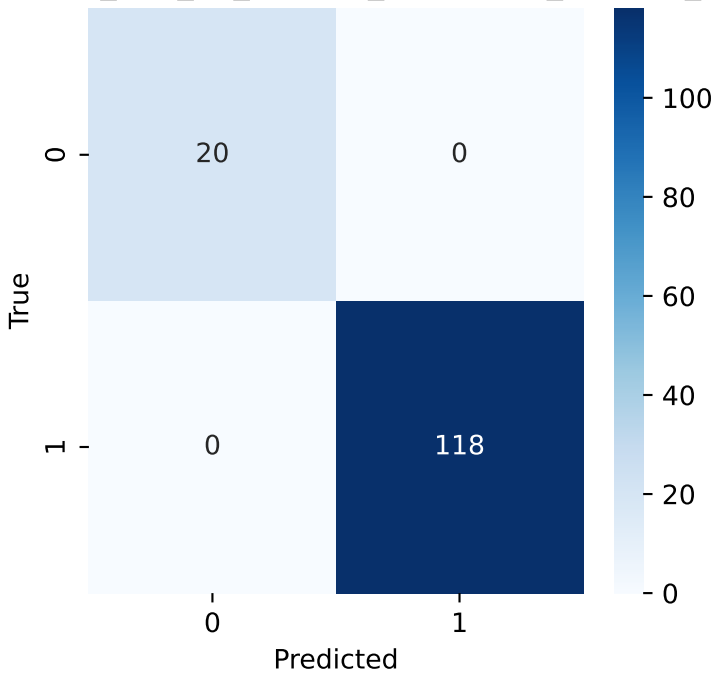
Matrix: abeg_1642_mi_session_20250630_1649



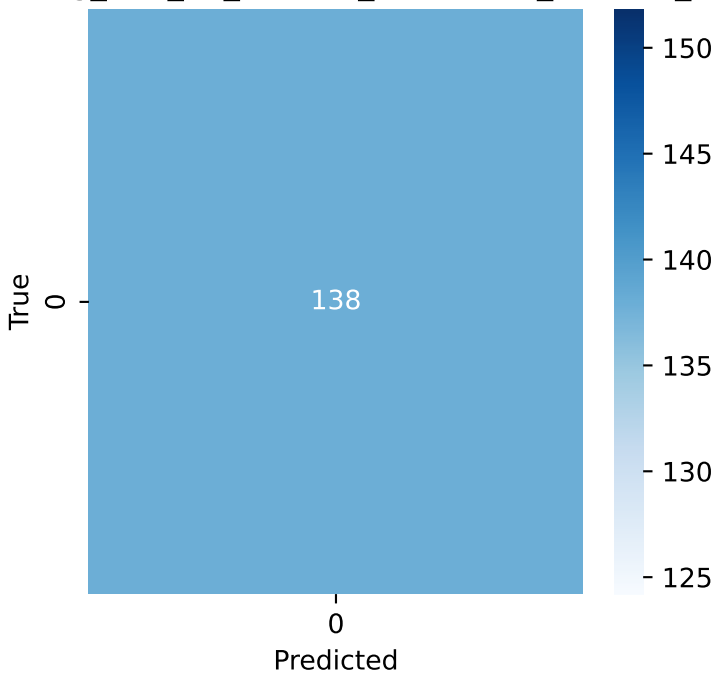
x: abeg_415_mi_session_20250701_041724_no



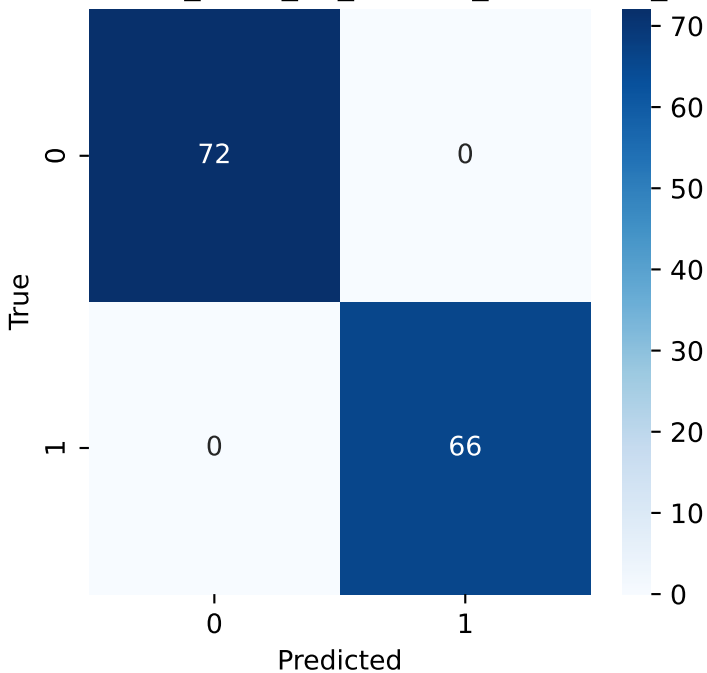
x: abeg_445_mi_session_20250701_044837_no



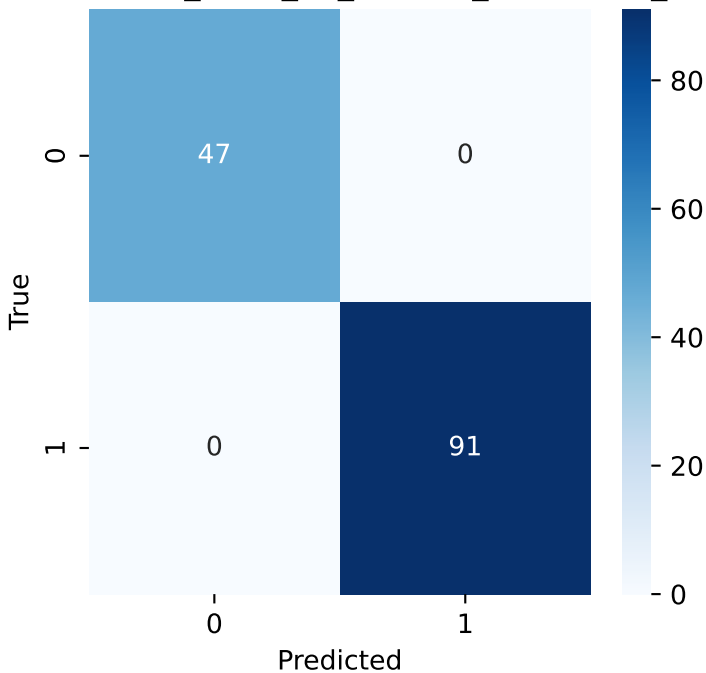
ix: abg_348_mi_session_20250701_035143_nor



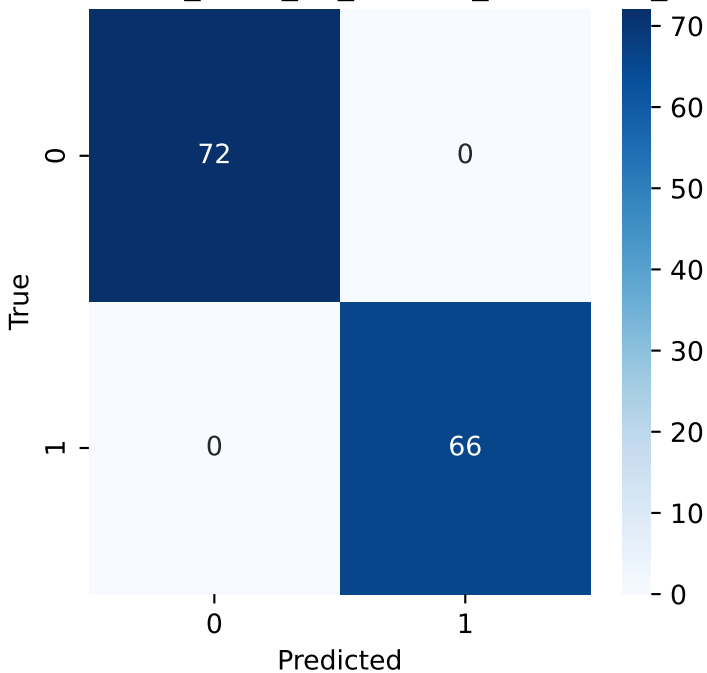
h Matrix: alex_2257_mi_session_20250626_2301



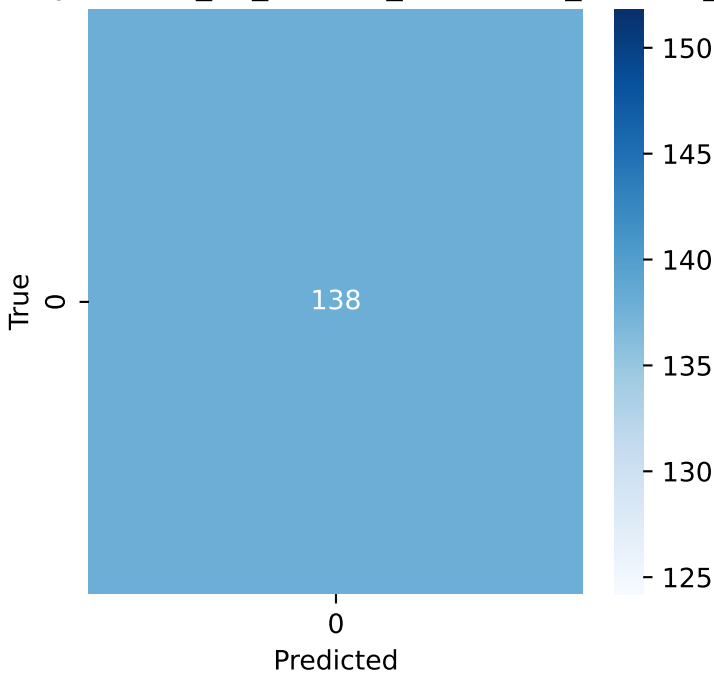
h Matrix: alex_2312_mi_session_20250626_2319



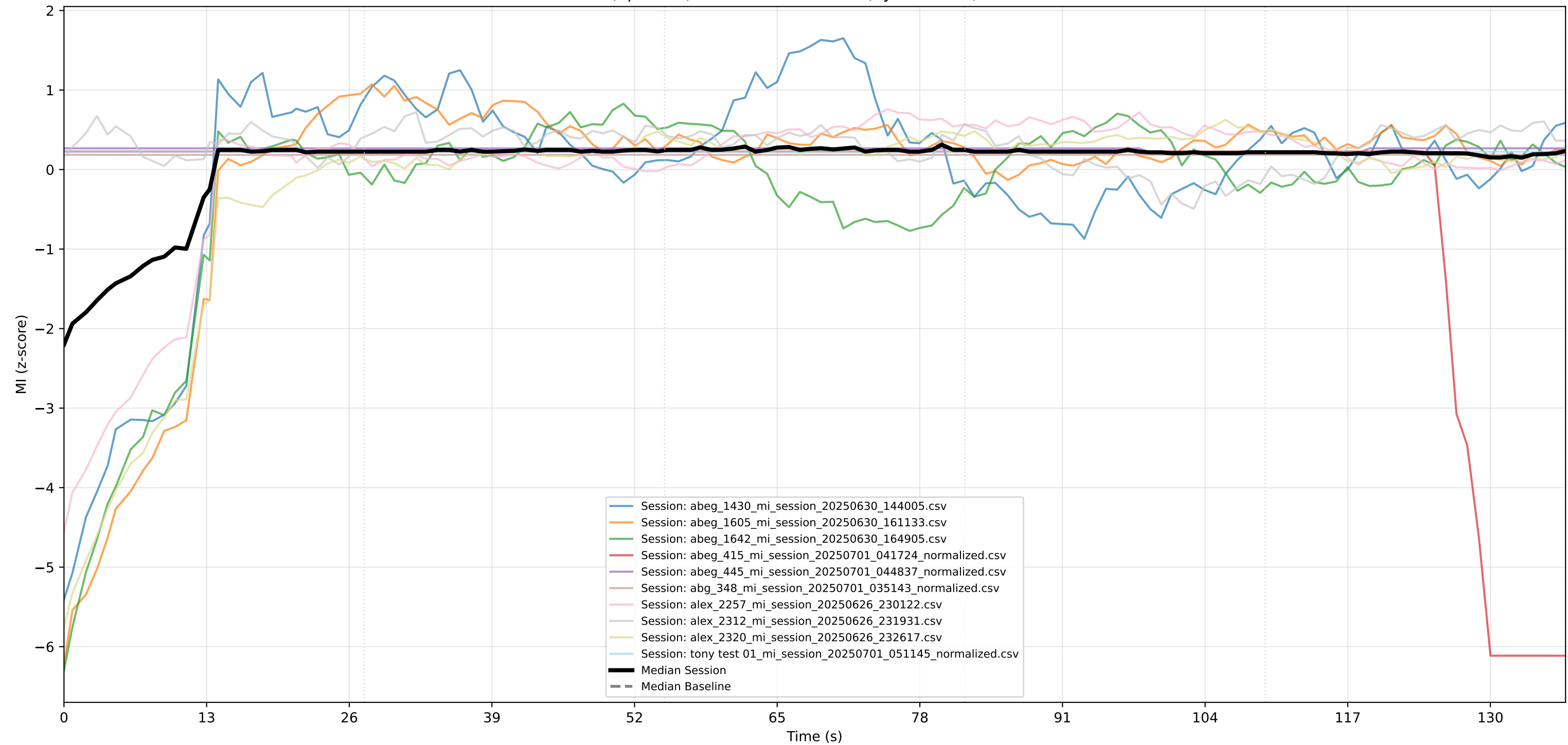
h Matrix: alex_2320_mi_session_20250626_2326



: tony test 01_mi_session_20250701_051145_no



Session MI (up to 10) vs. Median Baseline (Eyes Closed) - 137s Duration



MI Advanced Report

Generated: 20250717_115711

MI Advanced Report - Statistical Summary (1Hz Sampling)

1. Mindfulness Index (MI) Trajectory

- * All MI session files with ≥ 100 samples were included (100s minimum at 1Hz).
- * Individual and median MI curves were plotted with 20-second moving average (see Figure 8.1).
- * RM-ANOVA across six epochs (0-50s, 50-100s, 100-150s, 150-200s, 200-250s, 250-300s):
Anova

```
=====
      F Value Num DF  Den DF Pr > F
-----
Epoch  5.0816 5.0000 45.0000 0.0009
=====
```

- * Post-hoc pairwise comparisons (Tukey HSD):
Multiple Comparison of Means - Tukey HSD, FWER=0.05

```
=====
group1  group2 meandiff p-adj  lower  upper  reject
-----
  0-23s 114-137s   0.8101 0.0567 -0.0142 1.6343  False
  0-23s  23-46s   1.2022 0.0009  0.3779 2.0265   True
  0-23s  46-68s   1.1758 0.0013  0.3516 2.0001   True
  0-23s  68-91s   1.1087 0.0028  0.2844  1.933   True
  0-23s  91-114s   1.0729 0.0041  0.2487 1.8972   True
114-137s 23-46s   0.3921 0.7235 -0.4321 1.2164  False
114-137s  46-68s   0.3658 0.7777 -0.4585  1.19  False
114-137s  68-91s   0.2986 0.8909 -0.5256 1.1229  False
114-137s  91-114s   0.2629 0.9336 -0.5614 1.0871  False
 23-46s  46-68s  -0.0264  1.0 -0.8506 0.7979  False
 23-46s  68-91s  -0.0935 0.9994 -0.9178 0.7308  False
 23-46s  91-114s -0.1293 0.9972 -0.9535  0.695  False
 46-68s  68-91s  -0.0672 0.9999 -0.8914 0.7571  False
 46-68s  91-114s -0.1029 0.9991 -0.9272 0.7214  False
 68-91s  91-114s -0.0358  1.0  -0.86 0.7885  False
=====
```

- * Time-frequency analysis (FFT) of the median MI curve was performed (see Figure 8.3).

2. MI Dynamics

- * Peak analysis (median curve): 0 peaks ($z > 1.5$) at sample times [], amplitudes [], durations [] (see Table 8.2, Figure 8.4).
- * AUC (median): 11.46; mean $\pm$ SD (individual): 4.31 $\pm$ 25.87 (see Table 8.3).
- * Bootstrap 95% CI for mean MI difference (0-135s vs 136-270s): mean diff = 0.000, CI = [0.000, 0.000].
- * Change point detection (Pelt algorithm): 1 change points at sample indices [10] (see Figure 8.5).

3. Clustering & Mixed-Effects

- * K-Means clustering (k=2): cluster averages {}, user distribution {0: 9, 1: 1} (see Figure 8.6).

* Mixed-effects model (MI ~ time + (1|Subject)): Mixed Linear Model Regression Results

```
=====
Model:          MixedLM Dependent Variable: value
No. Observations: 1380  Method:          REML
No. Groups:      10     Scale:           1.0048
Min. group size:  138   Log-Likelihood: -1977.5717
Max. group size:  138   Converged:       Yes
Mean group size: 138.0
=====
      Coef.  Std.Err.  z    P>|z|  [0.025 0.975]
-----
Intercept  -0.307    0.081  -3.815 0.000  -0.465 -0.149
time       0.005    0.001  7.056 0.000  0.003  0.006
file Var   0.036    0.020
=====
```